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AIware: An Embedded AI System for Real-Time Driver State Monitoring and Alerting

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PCS3848 - Sistemas Embarcados

Universidade de São Paulo – USP
Escola Politécnica
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Resumo

Apresentamos o projeto do nosso grupo: AIware, um sistema de inteligência artificial embarcada para monitoramento e alerta do estado do condutor em tempo real. Nossa proposta de projeto visa enfrentar o problema dos acidentes de trânsito causados por fadiga e distração por meio de uma solução de baixo custo e alta eficiência. A proposta foca no estudo de trabalhos relacionados explorando modelos de Deep Learning compactos (TinyML) e como otimizá-los para hardware com recursos limitados

Palavras-chave: Embedded AI. TinyML. Driver Monitoring Systems (DMS).

Abstract

We present our group's project: AIware, an embedded artificial intelligence system for real-time driver state monitoring and alerting. Our project proposal aims to tackle the problem of traffic accidents caused by fatigue and distraction through a low-cost, high-efficiency solution. The proposal focuses on the study of related works exploring tiny Deep Learning models optimized for resource-constrained hardware.

Keywords: Embedded AI. TinyML. Driver Monitoring Systems (DMS).

Contents

1 INTRODUCTION 5
1.1 Context 5
1.2 Motivation 5
1.3 Objectives 6
1.4 Related works 6
1.5 Commercial Driver Monitoring Systems 8
1.6 Organisation 8

2 CAMERA-BASED DRIVER MONITORING 9
2.1 Near-Infrared (NIR) Imaging 9
2.2 Behavioral Biometrics: EAR and MAR Metrics 9
2.3 Remote Photoplethysmography (rPPG) 10

3 PROJECT REQUIREMENT SPECIFICATION 11

4 PROJECT ARCHITECTURE 13

5 CONCLUSIONS 15

BIBLIOGRAPHY 16

1 Introduction

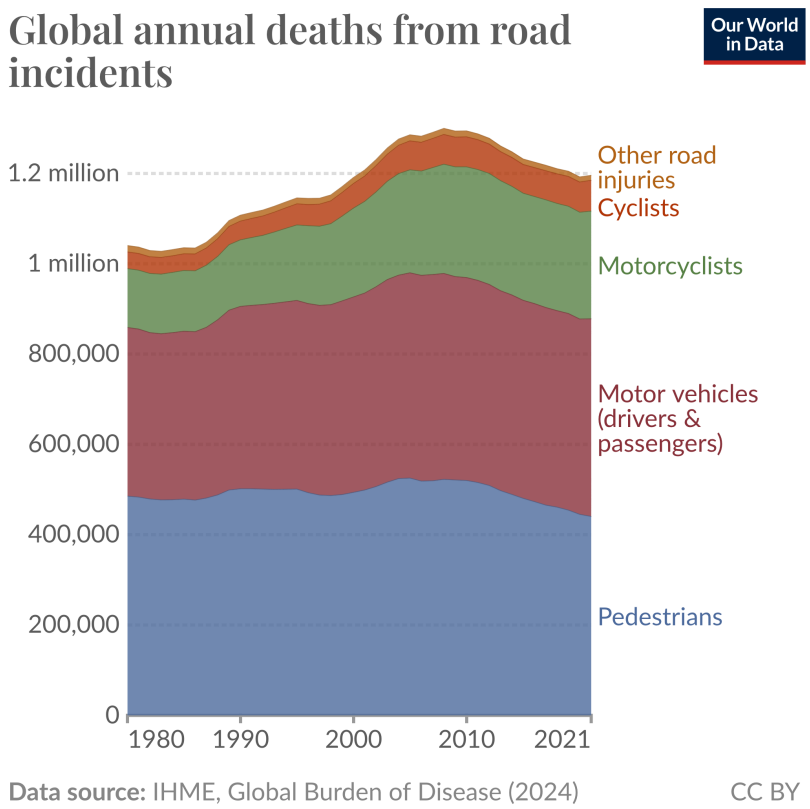
1.1 Context

This monograph documents a project proposal submitted for the course PCS3848: Embedded Systems at the Escola Politécnica da Universidade de São Paulo (EPUSP). It contains the specification for an embedded AI system for real-time driver state monitoring and alerting, to be implemented in the Embedded Systems Laboratory course.

1.2 Motivation

According to the World Health Organization (World Health Organization (WHO), 2023), approximately 1.19 million people die in road traffic accidents every year with the number of injuries reaching 50 million people per year. The main factors increasing the risk of injury in road traffic accidents include: speeding, driving under the influence of alcohol or drugs, non-use of seat-belts and child restraints and distracted driving.

Figure 1 – Number of deaths in traffic accidents worldwide (RITCHIE, 2024)



During our project discussions, we observed that these factors could be detected

by a smart computer vision system installed in the car's dashboard with a sufficiently powerful artificial intelligence model. After consulting related works in the literature (LI, 2024) (AICHA, 2025) (BOON et al., 2026) (ESSAHRAUI et al., 2026), we were convinced that such a project is viable with relatively modest hardware constraints and decided to propose it as our project for the course.

1.3 Objectives

We propose a real-time driver state monitoring system for cars and trucks.

The primary objective of this system is to help drivers, specially when they are alone in the vehicle for prolonged periods, by alerting and preventing them from driving in case they are speeding, under the effect of alcohol or other drugs, not wearing their seat-belt, distracted or sleepy.

All these traffic infractions are factors that contribute to potentially serious accidents, especially for truck drivers. The idea is to use simple computer vision models trained with specialized platforms such as TensorFlow Lite to capture images of the driver and, in the event an undesired state is detected, emit an audible alert and provide appropriate recommendations to assist the driver and engage their attention.

Observe that the key metrics to be minimized here are **latency per model inference** and **accuracy** so the drive can react sufficiently fast in the vast majority of risky scenarios. Unlike some embedded systems applications, power consumption is not critical here because the system can be powered via USB using the car's battery during its operation.

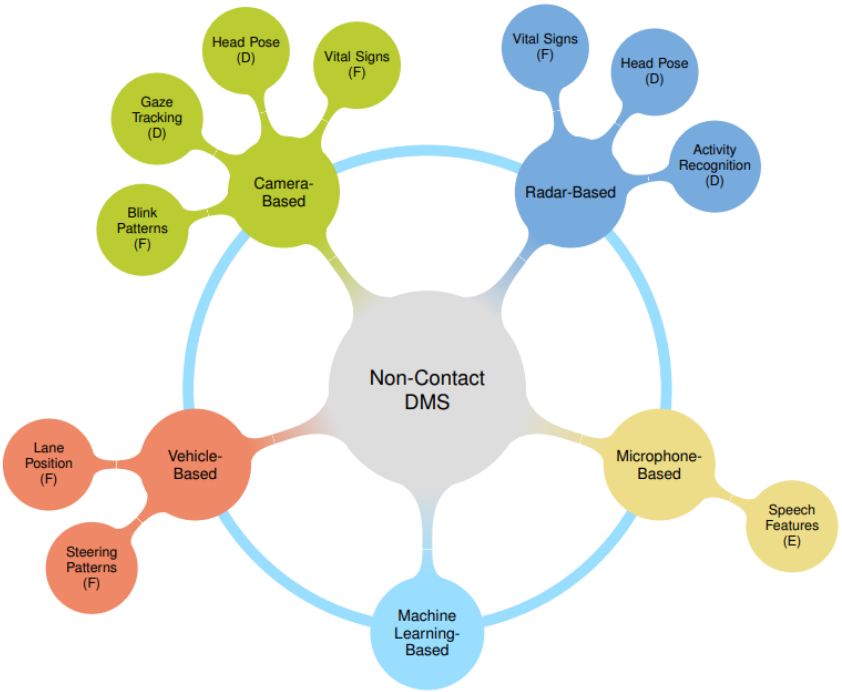
1.4 Related works

The development of driver monitoring systems is an active field of research (RAHMAN et al., 2026) involving many different types of sensors: cameras, microphones, radar, biometric sensors and even exploiting mechanical on the vehicle. There is a critical trade-off between sensing accuracy and the computational constraints of embedded hardware. While non-visual methods like radar offer privacy benefits, camera-based systems remain the most versatile for capturing nuanced behavioral indicators. For this reason, we choose to focus on the camera-based visual approach.

The most relevant works consulted are summarized below:

(a) (ESSAHRAUI et al., 2026) benchmarks seven different lightweight deep learning architectures for real-time embedded driver state detection. The benchmark uses Kaggle's State Farm Distracted Driver Detection multimodal classification dataset

Figure 2 – Different sensing technologies used in driver monitoring systems (RAHMAN et al., 2026)



to evaluate and compare the model architectures. The architectures considered achieve accuracies of over 90% with very small footprints, sometimes smaller than one megabyte.

(b) (BOON et al., 2026) presents a robust embedded solution implemented on NVIDIA's Jetson Orion Nano GPU using the YOLOv11 nano (YOLOv11n) model architecture for real-time drowsiness monitoring in drivers. The methodology focuses on high-speed object detection to monitor facial traits in the driver.The quantities monitored include Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) to identify states of drowsiness or distraction. The system achieves a low latency per inference of approximately 22ms with n overall accuracy of 93.1%.

Figure 3 – Performance for different tiny ML architectures on the State Farm Distracted Driver Detection dataset (ESSAHRAUI et al., 2026)

Model	Accuracy (%)	Loss	Precision (%)	Recall (%)	F1-score (%)	RAM (MB)	Speed (FPS)	Size (MB)
Efficient-Tiny	91.64	0.4243	93.02	92.56	92.01	0.08	26.81	0.43
ESPNetv2-Small	99.50	0.0482	99.45	99.44	99.44	0.22	40.49	0.74
MCUNet	97.85	0.0951	97.69	97.48	97.54	0.03	52.72	0.20
Micro-MobileNet	98.99	0.0300	99.26	99.23	99.24	0.15	37.71	0.60
PhiNets	99.66	0.0175	99.50	99.49	99.49	0.15	36.14	0.60
ShuffleNetLite	86.06	0.6538	92.93	86.23	87.21	1.75	22.27	5.46
SqueezeNetv11	95.81	0.1399	95.85	95.76	95.76	0.05	28.60	0.33

1.5 Commercial Driver Monitoring Systems

After our bibliographical review, we started looking for material on driver monitoring systems from companies to see what commercial products already exist or are being developed for the market. We soon realised that the field of Driver Monitoring Systems (DMS) is characterized by a significant divide between high-end, proprietary solutions and research-oriented embedded systems.

In the United States and Europe, there is movement to popularize DMS in a wide range of vehicles and legislation is likely to impose DMS soon, specially trucks where they could prevent most accidents. Startups like **Seeing Machines**, **Novus Hi-Tech** and **Mobileye** develop commercial DMS for big automotive companies like indian manufacturer **Eicher Motors**, **Tesla** and **Stellantis** with robust performance, but they often require expensive, specialized hardware and are integrated deeply into a vehicle's proprietary architecture.

In contrast, the embedded system architectures we saw in our bibliographical study are designed as a generic, modular, low-cost alternative that maintains competitive performance through optimized TinyML architectures. Our project, AIware, fits in this research trend. Instead of developing a system integrated with a manufacturer's proprietary technology, the goal of our system is to allow implantation on any vehicle after manufacture is finished.

1.6 Organisation

In chapter 2, we explore key concepts used by the driver monitoring systems discussed. Then, in chapter 3, we list functional and non-functional requirements for the project. In chapter 4 we provide an architecture for the project detailing what components could be used and how to integrate them into the system. Finally, in chapter 5 we present our conclusions and final considerations ahead of the Laboratory course where we intend to implement this system.

2 Camera-Based Driver Monitoring

In this chapter, we introduce some interesting concepts analysed by the group when studying literature for camera-based driver monitoring systems.

2.1 Near-Infrared (NIR) Imaging

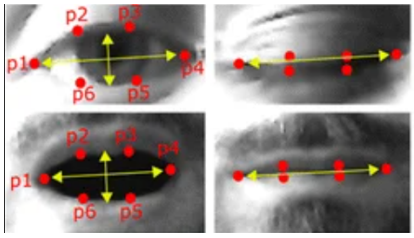
One of the primary challenges faced in the camera-based approach is maintaining consistent performance under varying lighting conditions, such as driving at night or through tunnels. According to (RAHMAN et al., 2026), Near-Infrared (NIR) imaging is the standard solution to this problem. Unlike standard RGB cameras, NIR systems operating in the $850 - -940$ nanometer can capture facial features in total darkness. In our AIware project, if using an infrared camera is not possible, we can try to incorporate microphone-based based to help our model make good predictions even when image quality is suboptimal.

2.2 Behavioral Biometrics: EAR and MAR Metrics

To objectively quantify driver states like drowsiness and distraction, two of the most critical metrics are the Eye Aspect Ratio (EAR) and the Mouth Aspect Ratio (MAR).

- **EAR** monitors the distance between eyelids to detect blinks and prolonged eye closure (micro-sleeps).
- **MAR** measures mouth opening levels to identify frequent yawning, a primary indicator of early-stage fatigue.

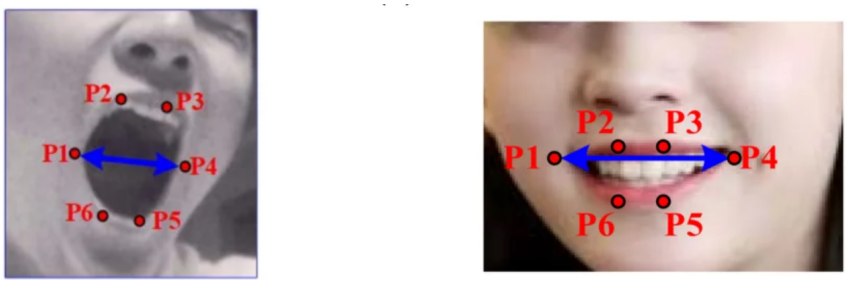
Figure 4 – Eye Aspect Ratio metric (BELLINO, 2018)



$$\mathrm{EAR} = \mathrm{MAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$

According to (RAHMAN et al., 2026), these geometric ratios allow lightweight AI models to perform binary classifications with high precision and low computational overhead, fitting the constraints of edge devices.

Figure 5 – Mouth Aspect Ratio metric (ZHU et al., 2022)



2.3 Remote Photoplethysmography (rPPG)

A more advanced concept discussed in recent literature is remote Photoplethysmography (rPPG), which allows for the detection of physiological signals such as heart rate and respiration using only a standard camera sensor. By analyzing minute skin color variations caused by blood circulation (which are often imperceptible to the human eye), the system can infer the driver's stress levels or sudden health emergencies. This is a very interesting technique, however it is unlike we will explore it for the AIware project in the Embedded Systems Laboratory course.

3 Project Requirement Specification

Functional Requirements

FR1 Image capture and processing must occur in real-time.

FR2 AI model inferences must be performed locally on the microcontroller/processor.

FR3 The entire system must be capable of performing offline inferences.

FR4 The camera must capture clear images during both day and night conditions.

FR5 The camera must feature infrared (IR) sensitivity.

FR6 The camera must be equipped with a microphone.

FR7 The system must encapsulate physical image storage devices within a "black box" enclosure.

FR8 The system must operate only when the vehicle is in motion.

FR9 The system must be equipped with an accelerometer to detect vehicle movement.

FR10 The system must alert the driver of inappropriate behavior through audible and visual signals.

FR11 The system must transmit the driver's state and history to the central hub whenever an internet connection is available.

FR12 The system must detect driver drowsiness.

FR13 The system must detect if the driver is using a mobile phone while driving.

FR14 The system must detect if the driver is distracted by other activities (e.g., one hand off the steering wheel or general inattention).

FR15 The system must detect if the driver is emotionally compromised (e.g., extreme sadness or anger).

FR16 The system must detect if the driver is intoxicated.

FR17 The system must detect if the vehicle has been involved in an accident.

FR18 In the event of an accident, the system must send an alert signal with the vehicle's location to the central hub.

FR19 While connected to a network, the system must periodically send a signal to the central hub for user position monitoring.

FR20 The system must encrypt all images captured and stored in the black box.

FR21 Only the central hub shall have the authority and means to decrypt the images stored in the black box.

Non-Functional Requirements

NFR1 Driver state classification must be completed in less than 100ms.

NFR2 The camera resolution must be at least 2 MP.

NFR3 The camera must support a minimum frame rate of 30 FPS.

NFR4 The black box enclosure must withstand vehicle collisions with forces up to 100 kN.

NFR5 The black box storage must have the capacity to store up to 12 hours of video footage.

NFR6 The luminous intensity of the visual alert signal must be between 500 and 1000 nits to ensure visibility without compromising the driver's vision sensitivity.

NFR7 The sound intensity of the audible alert signal must exceed 70 dB.

NFR8 When an internet connection is available, the system must transmit its location and the driver's state to the central hub at intervals of no more than 30 seconds.

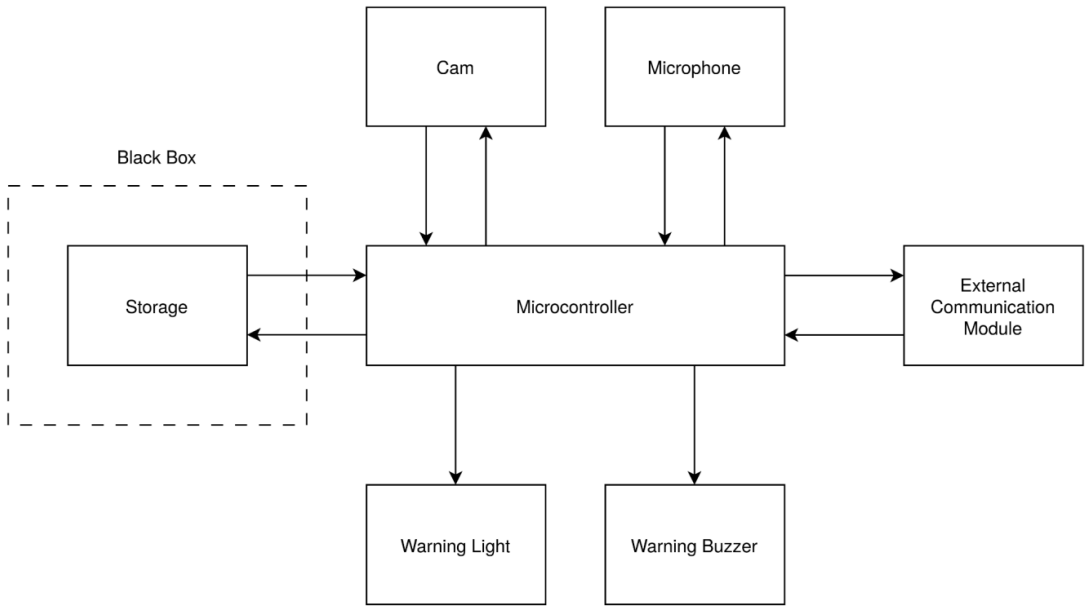
NFR9 The system must maintain a cumulative availability of 99%.

NFR10 The driver state inference model must achieve an accuracy higher than 95%.

4 Project Architecture

In this chapter, we translate the listed requirements into a selection of specific components to constitute the final system. The primary platform for implementing the artificial intelligence models will be a Raspberry Pi 3B+. The device features 1GB of RAM, which is expected to be sufficient for executing lightweight AI models following preliminary image processing. For prototyping purposes, an OV7670 camera with 300KP resolution (currently owned by a member of the group) may be used; however, if available in the laboratory the final product is intended to utilize a camera with a minimum resolution of 2MP.

Figure 6 – Block diagram with the AIware system architecture



To generate the audible alert, a standard 5V active buzzer will be employed, as it is capable of emitting approximately 85dB at a distance of 10cm, ensuring effectiveness given the system's proximity to the driver. Regarding the visual alert signal, 5mm LEDs are considered for the prototyping phase, with the potential transition to a high-intensity light source for the final version to meet the required luminance specifications.

Regarding external communication, the system implements the MQTT (Message Queuing Telemetry Transport) protocol to transmit driver state and telemetry data to the central monitoring hub. By using MQTT, the AIware system can send periodic updates with minimal protocol overhead, reducing both latency and bandwidth consumption. Furthermore, the protocol's Quality of Service (QoS) levels allow the system to prioritize critical event notifications such as the detection of severe fatigue or a medical emergency ensuring that these alerts are acknowledged by the central server even if the connection is

momentarily interrupted.

Initially, LoRa was considered as a possible alternative in our group discussions because of the low volume, low energy consumption and because LoRa stations can cover big distances. However, we opted for MQTT in the end because LoRa requires a dedicated network of gateways. Since a vehicle is mobile and may travel across cities or highways, relying on a provider's private LoRa network is impractical. Using the existing Cellular Network via MQTT seems like a simpler and more straightforward solution.

5 Conclusions

In this monograph we presented the activities we developed in the course and some of the study materials we reviewed. With this project proposal for the Alware system, we studied topics in the intersection of safety analysis and embedded AI.

The literature review provided an encouraging outlook for the project's feasibility. Using TinyML architectures, high-accuracy driver state detection is achievable even on resource-constrained hardware (ESSAHRAUI et al., 2026) (BOON et al., 2026).

A significant portion of this study was dedicated to the project requirement specification and architecture. Moving forward into the Embedded Systems Laboratory course, our focus will shift from theoretical specification to practical development. This will include model training and optimization using TensorFlow Lite to compress deep learning models for the Raspberry Pi environment; physical integration of the camera to ensure good quality a real-time image data stream; and assembling the black box enclosure, ensuring the buzzer and LED alerts meet the specified intensity requirements for driver engagement.

In summary, while this is an initial study, the clarity of the requirements and the efficiency of the proposed architecture suggest that Alware is a feasible solution for enhancing driver safety through low-cost, high-efficiency embedded AI.

Bibliography

AICHA, F. B. *Optimized embedded AI: efficient implementation of CNNs on ESP32-CAM for real-time image classification*. 2025. Springer-Verlag GmbH Austria. Disponível em: <https://doi.org/10.1007/s00607-025-01559-z>. Citado na página 6.

BELLINO, A. Sequence: a remote control technique to select objects by matching their rhythm. *Personal and Ubiquitous Computing*, v. 22, 08 2018. Citado na página 9.

BOON, E. L. et al. *Lumina Alert: A YOLOv11n-Based Embedded System for Real-Time Driver Drowsiness Detection and Multi-Modal Intervention*. 2026. International Journal of Innovative Science and Research Technology. Disponível em: <https://www.researchgate.net/profile/Grazielle-Nychole-Dela-Cruz/publication/401350914_Lumina_Alert_A_YOLOv11n-Based_Embedded_System_for_Real-Time_Driver_Drowsiness_Detection_and_Multi-Modal_Intervention/links/69afdb1ba91b826e4348be2c/Lumina-Alert-A-YOLOv11n-Based-Embedded-System-for-Real-Time-Driver-Drowsiness-Detection-an pdf>. Citado 3 vezes nas páginas 6, 7, and 15.

ESSAHRAUI, S. et al. Tiny deep learning models for real-time and efficient embedded driver state detection. *Journal of Real-Time Image Processing*, Springer-Verlag GmbH Germany, v. 23, n. 39, 2026. Disponível em: <https://doi.org/10.1007/s11554-025-01836-2>. Citado 3 vezes nas páginas 6, 7, and 15.

LI, B. *Embedded Artificial Intelligence: Principles, Platforms and Practices*. Singapore: Springer Nature Singapore, 2024. ISBN 978-981-97-5037-5. Disponível em: <https://link.springer.com/book/10.1007/978-981-97-5038-2>. Citado na página 6.

RAHMAN, A. et al. Driver monitoring systems: Advances, challenges, and future directions in noncontact sensor technologies: A review. *IEEE Sensors Journal*, v. 26, n. 3, p. 3540–3557, 2026. Citado 3 vezes nas páginas 6, 7, and 9.

RITCHIE, H. *More than a million people die from road injuries every year*. 2024. Our World in Data. Disponível em: <https://ourworldindata.org/data-insights/more-than-a-million-people-die-from-road-injuries-every-year>. Citado na página 5.

World Health Organization (WHO). *Road traffic injuries*. 2023. Disponível em: <https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries>. Citado na página 5.

ZHU, T. et al. Research on a real-time driver fatigue detection algorithm based on facial video sequences. *Applied Sciences*, v. 12, p. 2224, 02 2022. Citado na página 10.